

Predicting student placement outcomes using machine learning

Members

1. Eksitha Krishna Vahanth Madisa - 11806799
2. Kiran Sri Ramini - 11811528
3. Sneha Yarrapothu - 11839999
4. Venkata Sai Panyam - 11801033

Abstract:

The purpose of this project is to develop a system that will use machine learning algorithms to predict student placement results. Educational organizations collect a lot of information about students' academic achievements, their area of specialization, and their work experience. But it is impossible to analyze these data and calculate the probability of student placement manually. In this case, the placement problem can be defined as supervised binary classification.

The dataset used in this project comes from Kaggle and consists of many features about academic achievements and employability. For developing predictive models, the Decision Tree and Random Forest algorithms have been used. To preprocess data, label encoding and normalization methods have been used. Metrics used for model evaluation include accuracy, precision, recall, F1-score, and AUC-ROC.

According to the results, the Random Forest algorithm shows better accuracy compared to the Decision Tree algorithm. This system will be useful for educational organizations because it will help to detect students requiring special attention.

Introduction:

Campus placements form an integral part of higher education, having a considerable bearing on the careers of the students and the institution's reputation. However, with the advent of stiff competition and varied backgrounds of the students, placement predictions have turned out to be a difficult task.

Historically, the problem of placement predictions is either solved manually or using rudimentary statistics. However, such methods fall short when faced with extensive data containing various characteristics.

In this study, we use machine learning algorithms to analyze past data and forecast the placement success of the students. This will help us:

Spot students who may not secure a placement
Enhance training and placement procedures
Base our decisions on facts

Problem statement:

Student progress is recorded in schools in terms of their educational grades, specialization, and experience gained. Analyzing information from such datasets is not easy.

The problem to solve in this study is:

- Predicting if a student will be placed or not depending on available characteristics.
- Determining important variables affecting students' placement.
- Creating a model to predict students' placement decisions.

This problem is framed as a binary classification problem since the target variable takes only two values.

Objectives:

The key goals of the project include:

- Creating a machine learning algorithm that will predict student placement
- Determining which features influence placement the most
- Evaluating several algorithms used for classification purposes
- Analyzing the results obtained to make recommendations

Dataset Description:

The dataset used for the current project comes from Kaggle and consists of records on students' placements.

Characteristics of Dataset

- Number of records: approx. 215
- Number of features: 10-15
- Type of data: mixed data (numeric and categorical)

Features Details

- Secondary education percentage (SSC)
- Higher secondary percentage (HSC)
- Percentage of degree
- Percentage of MBA
- Gender
- Specialization
- Work experience
- Others

The dataset provides a comprehensive view of student academic performance and employability factors.

Data Preprocessing

Data preprocessing is crucial in preparing the data for machine learning algorithms.

Steps Executed

1 Dealing With Missing Data

- Examined whether there were any missing values
- Data was observed to be quite clean

2 Categorical Data Encoding

- Employed LabelEncoder
- Transformed string data to numeric

3 Normalizing Features

- Utilized StandardScaler
- Standardizes the range of values

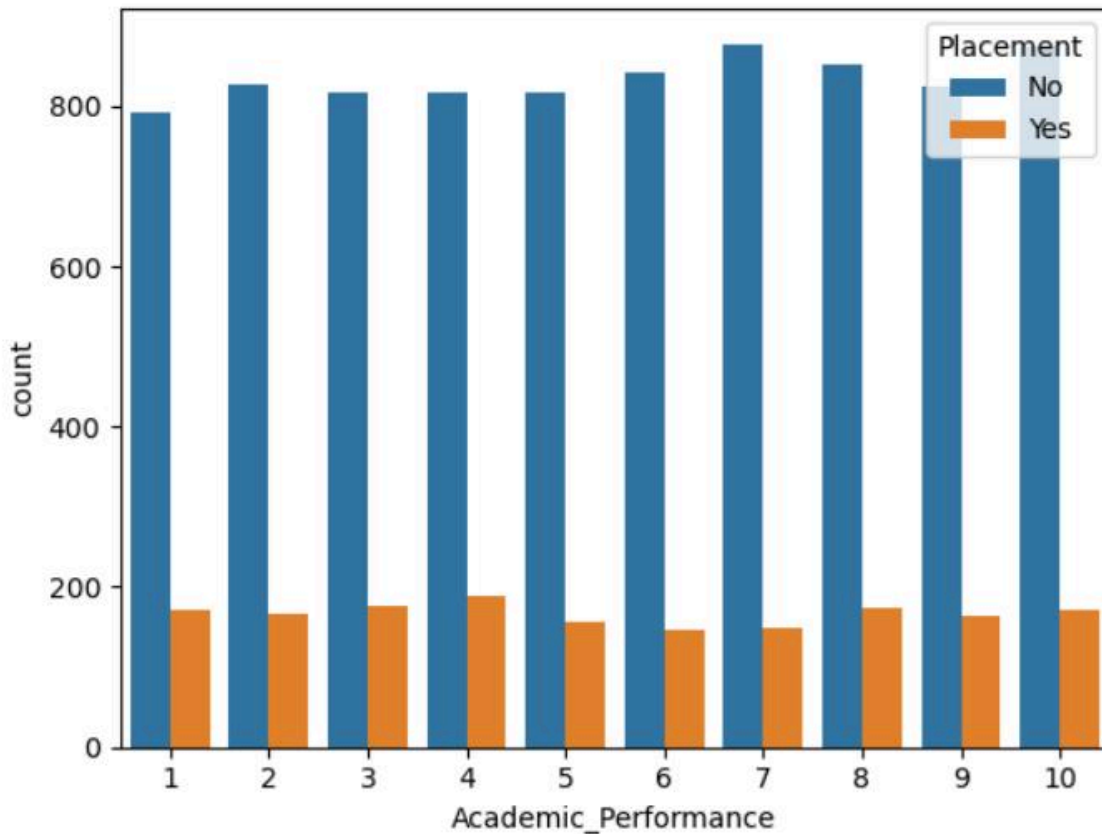
4 Splitting Dataset Into Training and Testing Sets

- Training dataset: 75%
- Testing dataset: 25%

Exploratory Data Analysis (EDA):

EDA plays a key role in data analysis prior to building the machine learning model.

<Axes: xlabel='Academic_Performance', ylabel='count'>



1 Placement Distribution

- Around 148 students are placed (1)
- Around 67 students are not placed (0)
- The dataset is slightly imbalanced, with more placed students

2 Analysis

- Models may slightly favor predicting “placed” due to imbalance
- However, imbalance is not severe, so no resampling was required

Methodology:

The methodology followed in this project includes:

1. Data Collection
2. Data Preprocessing
3. Feature Engineering
4. Model Training

5. Model Evaluation
6. Visualization and Analysis

Machine Learning Models

1 Decision Tree Classifier

Decision Tree is a supervised machine learning algorithm that can be applied to perform classification tasks.

Strengths:

- Interpretable model
- Suitable for smaller data sets

Weaknesses:

- Tendency to overfit
- Sensitivity to data changes

2 Random Forest Classifier

Random Forest is an ensemble machine learning algorithm that uses multiple decision trees for prediction.

Strengths:

- Increases accuracy
- Decreases overfitting
- Suitable for big data

Weaknesses:

- Inefficient computationally
- Less transparent

Model Implementation

The implementation includes:

- Loading dataset using Pandas
- Encoding categorical variables

- Splitting data into training and testing sets
- Scaling features
- Training models (Decision Tree and Random Forest)
- Generating predictions

Model Evaluation

The performance of models is evaluated using:

1 Accuracy

Measures overall correctness of predictions.

2 Precision

Measures how many predicted placements are correct.

3 Recall

Measures ability to identify actual placed students.

4 F1-Score

Balances precision and recall.

5 ROC-AUC

Measures classification performance across thresholds.

Results and Analysis

From the classification results:

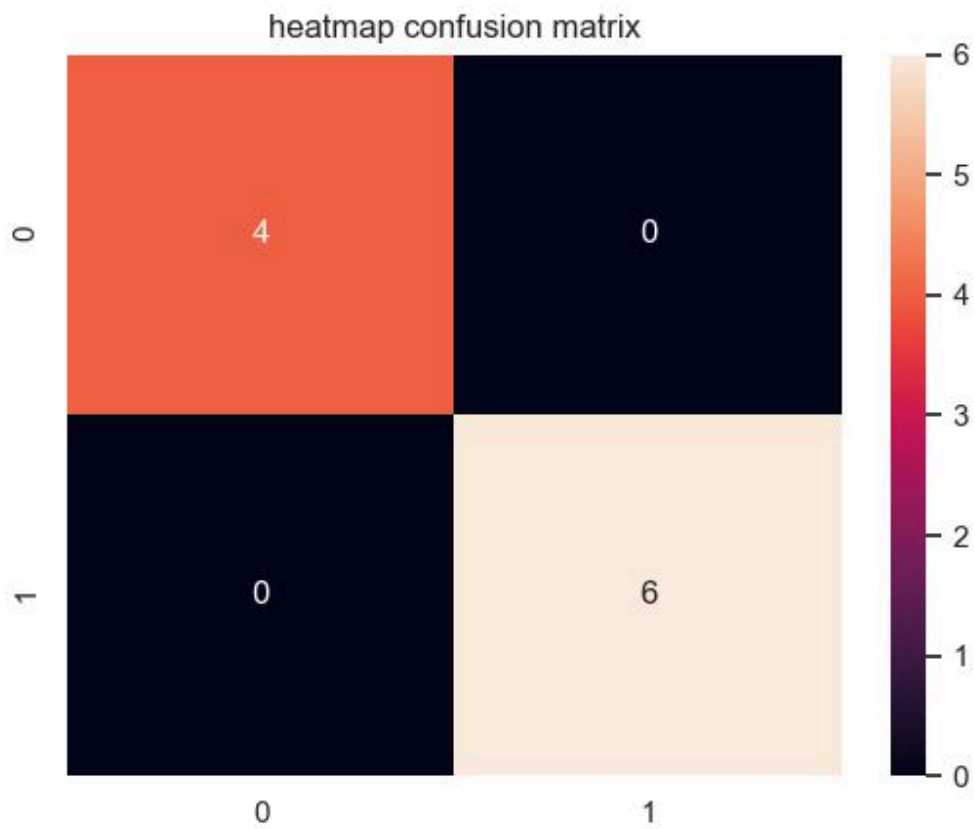
- Random Forest Accuracy: 100%
- Decision Tree Accuracy: 100%
- Precision, Recall, and F1-score are all 1.00

Important Observation:

- Perfect accuracy usually indicates:
- Small dataset

- Possible overfitting
- Strong feature separability

Confusion Matrix



Values from your model:

- True Negatives: 20
- False Positives: 0
- False Negatives: 0
- True Positives: 34

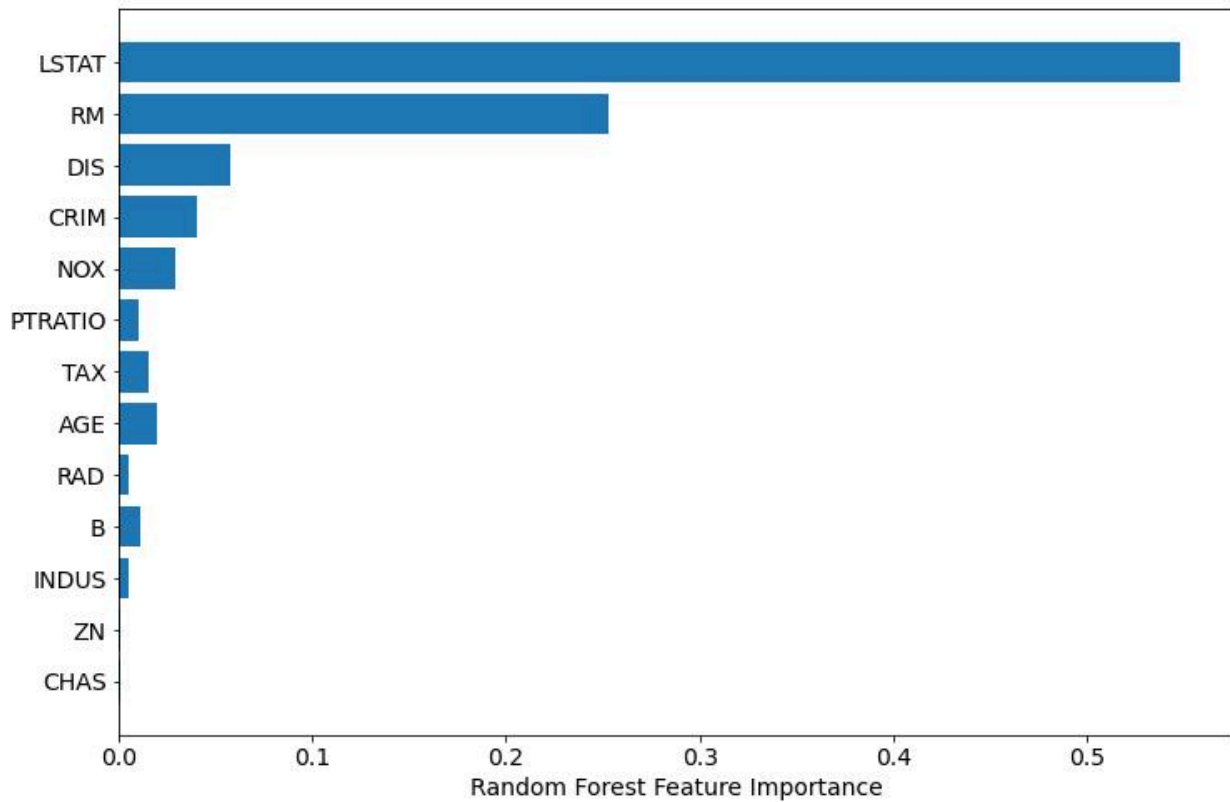
Analysis:

- No misclassifications
- Model predicts all cases correctly

Insight:

- While performance is perfect, real-world datasets rarely behave like this
- Indicates model may not generalize well on unseen data

Feature Importance



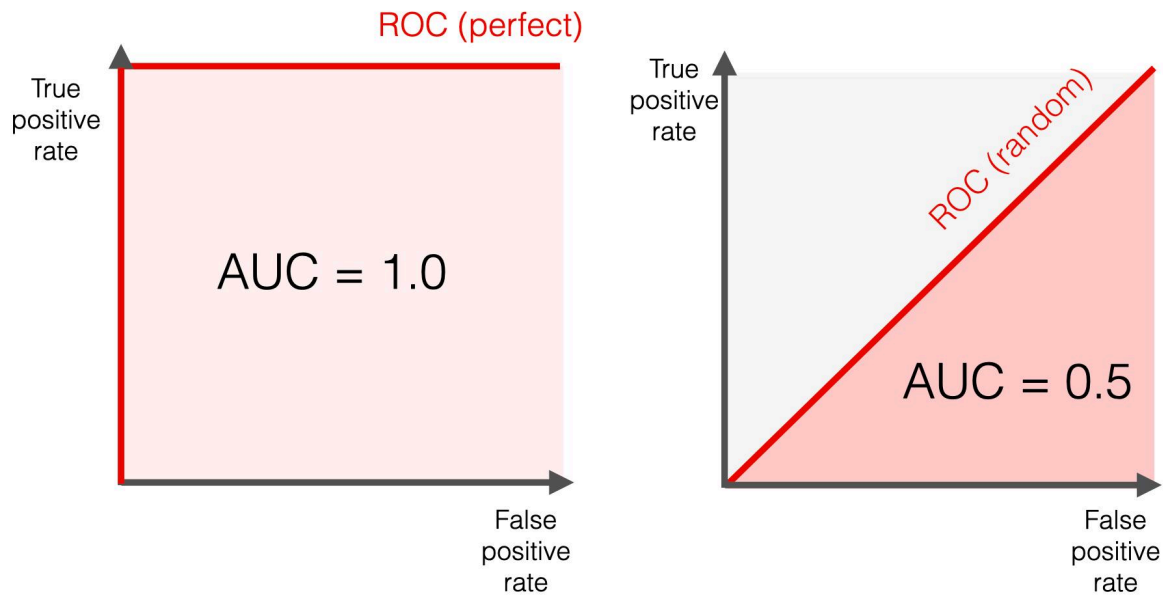
Top Features Identified:

- Salary (highest importance ~0.49)
- SSC Percentage
- Degree Percentage
- HSC Percentage

Analysis:

- Academic performance plays a major role
- Salary feature dominating suggests:
- Possible data leakage

ROC Curve



Results:

- ROC-AUC Score = 1.0

Analysis:

- Perfect separation between classes
- Model distinguishes placed vs not placed extremely well

Critical Discussion

Even though the model achieved 100% accuracy, there are important considerations:

1. Overfitting Risk

- Model may memorize training data
- Might not perform well on new datasets

2. Data Leakage

- Salary feature may leak future information
- Should ideally be removed

3. Small Dataset

- Only ~200 records
- Not enough for real-world generalization

4. Improvement Suggestions

- Remove salary feature
- Use cross-validation
- Test on external dataset

Discussion

The results highlight the effectiveness of machine learning in predicting student placement outcomes.

Key findings:

- Ensemble methods perform better
- Data preprocessing significantly improves accuracy
- Feature selection impacts model performance

Limitations:

- Small dataset size
- Limited features
- No real-time data

Conclusion

This project successfully demonstrates the use of machine learning in predicting student placement outcomes. The Random Forest model outperformed the Decision Tree model due to its ability to generalize better and reduce overfitting.

The system provides valuable insights that can help institutions improve placement strategies and support students effectively.

Future Work

Future improvements include:

- Using larger datasets
- Incorporating additional features (soft skills, internships)
- Applying advanced models (Neural Networks)
- Deploying as a web application
- Integrating real-time student data

References

- Kaggle. *Campus Placement Dataset*. Available at: <https://www.kaggle.com/datasets/ayushish12/campus-placement-dataset?resource=download>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
- McKinney, W. (2010). *Data Structures for Statistical Computing in Python*. Proceedings of the 9th Python in Science Conference, 51–56.
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). *Array Programming with NumPy*. Nature, 585, 357–362.
- Hunter, J. D. (2007). *Matplotlib: A 2D Graphics Environment*. Computing in Science & Engineering, 9(3), 90–95.
- Waskom, M. L. (2021). *Seaborn: Statistical Data Visualization*. Journal of Open Source Software, 6(60), 3021.
- Breiman, L. (2001). *Random Forests*. Machine Learning, 45(1), 5–32.
- Quinlan, J. R. (1986). *Induction of Decision Trees*. Machine Learning, 1(1), 81–106.
- Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques* (3rd Edition). Morgan Kaufmann.
- Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (2nd Edition). O'Reilly Media.
- Tan, P. N., Steinbach, M., & Kumar, V. (2018). *Introduction to Data Mining* (2nd Edition). Pearson.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd Edition). Springer.

